

# Introduction to Computer Organisation and Architecture UCCD1133



## Chapter 2- Part I Number systems and Data Representation

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# Outline

- Number Bases
- Computer Code
- Representation of negative numbers

# Number Bases

- Decimal (Base 10) = Base (radix, R)
  - Digits {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
  - R = 10.
- Binary (Base 2)
  - Digits {0, 1}
  - R = 2.
- Octal (Base 8)
  - Digits {0, 1, 2, 3, 4, 5, 6, 7}
  - R = 8.
- Hexadecimal (Base 16) 0~F
  - Digits {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F}
  - R = 16.

| Decimal | Binary | Octal | Hexadecimal |
|---------|--------|-------|-------------|
| 0       | 0      | 0     | 0           |
| 1       | 1      | 1     | 1           |
| 2       | 10     | 2     | 2           |
| 3       | 11     | 3     | 3           |
| 4       | 100    | 4     | 4           |
| 5       | 101    | 5     | 5           |
| 6       | 110    | 6     | 6           |
| 7       | 111    | 7     | 7           |
| 8       | 1000   | 10    | 8           |
| 9       | 1001   | 11    | 9           |
| 10      | 1010   | 12    | a           |
| 11      | 1011   | 13    | b           |
| 12      | 1100   | 14    | c           |
| 13      | 1101   | 15    | d           |
| 14      | 1110   | 16    | e           |
| 15      | 1111   | 17    | f           |
| 16      | 10000  | 20    | 10          |

# Binary Number

- The binary number system has two digits:
  - 0 represents the no voltage or an off state
  - 1 represents the presence of voltage or an on state
- The binary numbering system has a base of 2 with each position weighted by a factor of 2:

| POSITIVE POWERS OF TWO<br>(WHOLE NUMBERS) |       |       |       |       |       |       |       |       | NEGATIVE POWERS OF TWO<br>(FRACTIONAL NUMBER) |          |          |          |          |          |
|-------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-----------------------------------------------|----------|----------|----------|----------|----------|
| $2^8$                                     | $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ | $2^{-1}$                                      | $2^{-2}$ | $2^{-3}$ | $2^{-4}$ | $2^{-5}$ | $2^{-6}$ |
| 256                                       | 128   | 64    | 32    | 16    | 8     | 4     | 2     | 1     | 1/2                                           | 1/4      | 1/8      | 1/16     | 1/32     | 1/64     |
|                                           |       |       |       |       |       |       |       |       | 0.5                                           | 0.25     | 0.125    | 0.0625   | 0.03125  | 0.015625 |

# HEXADECIMAL NUMBERS

## BASE 16

- The Hexadecimal (HEX) number system has a base of sixteen as shown in the Table.
- It is composed of 16 numeric and alphabetic characters
- Hexadecimal digit represent a 4-bit binary number.

| DECIMAL | BINARY | HEX |
|---------|--------|-----|
| 0       | 0000   | 0   |
| 1       | 0001   | 1   |
| 2       | 0010   | 2   |
| 3       | 0011   | 3   |
| 4       | 0100   | 4   |
| 5       | 0101   | 5   |
| 6       | 0110   | 6   |
| 7       | 0111   | 7   |
| 8       | 1000   | 8   |
| 9       | 1001   | 9   |
| 10      | 1010   | A   |
| 11      | 1011   | B   |
| 12      | 1100   | C   |
| 13      | 1101   | D   |
| 14      | 1110   | E   |
| 15      | 1111   | F   |

# OCTAL NUMBERS

## BASE 8

- The Octal number system is composed of 8 digits, which are; 0,1,2,3,4,5,6,7 as shown in table below

| Decimal | Binary | Octal |
|---------|--------|-------|
| 0       | 000    | 0     |
| 1       | 001    | 1     |
| 2       | 010    | 2     |
| 3       | 011    | 3     |
| 4       | 100    | 4     |
| 5       | 101    | 5     |
| 6       | 110    | 6     |
| 7       | 111    | 7     |

# Convert Base N to Decimal (Base 10)

Example 1: Convert the binary whole number 1101101 to decimal

The decimal value of any binary number can be found by adding the weights of all bits that are 1 and discarding the weights of all bits that are 0.

|               |       |       |       |       |       |       |       |
|---------------|-------|-------|-------|-------|-------|-------|-------|
| Weight        | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
| Binary Number | 1     | 1     | 0     | 1     | 1     | 0     | 1     |

- Therefore  $1101101 = 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^0$   
 $= 64 + 32 + 8 + 4 + 1$   
 $= 109_{10}$

Weights in a binary number are based on power of two, that increase from right to left, beginning with  $2^0$

# Convert Base N to Decimal (Base 10)

Example 2 :Convert the fractional binary number 110.1011 to decimal

|               |       |       |       |   |          |          |          |          |
|---------------|-------|-------|-------|---|----------|----------|----------|----------|
| Weight        | $2^2$ | $2^1$ | $2^0$ | . | $2^{-1}$ | $2^{-2}$ | $2^{-3}$ | $2^{-4}$ |
| Binary Number | 1     | 1     | 0     | . | 1        | 0        | 1        | 1        |

- Therefore  $110.1011 = 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^{-1} + 1 \times 2^{-3} + 1 \times 2^{-4}$   
 $= 4 + 2 + 0.5 + 0.125 + 0.0625$   
 $= 6.6875_{10}$

# Convert Base N to Decimal (Base 10)

Example 3 : Convert the hexadecimal number to decimal

a)  $BE_{16}$

b)  $54_{16}$

Answer

a)  $BE_{16} =$   
 $B \times 16^1 + E \times 16^0 = 11 \times 16 + 14 \times 1 = 190$

b)  $54_{16}$   
 $5 \times 16^1 + 4 \times 16^0 = 5 \times 16 + 4 \times 1 = 84$

Example 4 : Convert the octal number to decimal

a)  $124_8$

b)  $276_8$

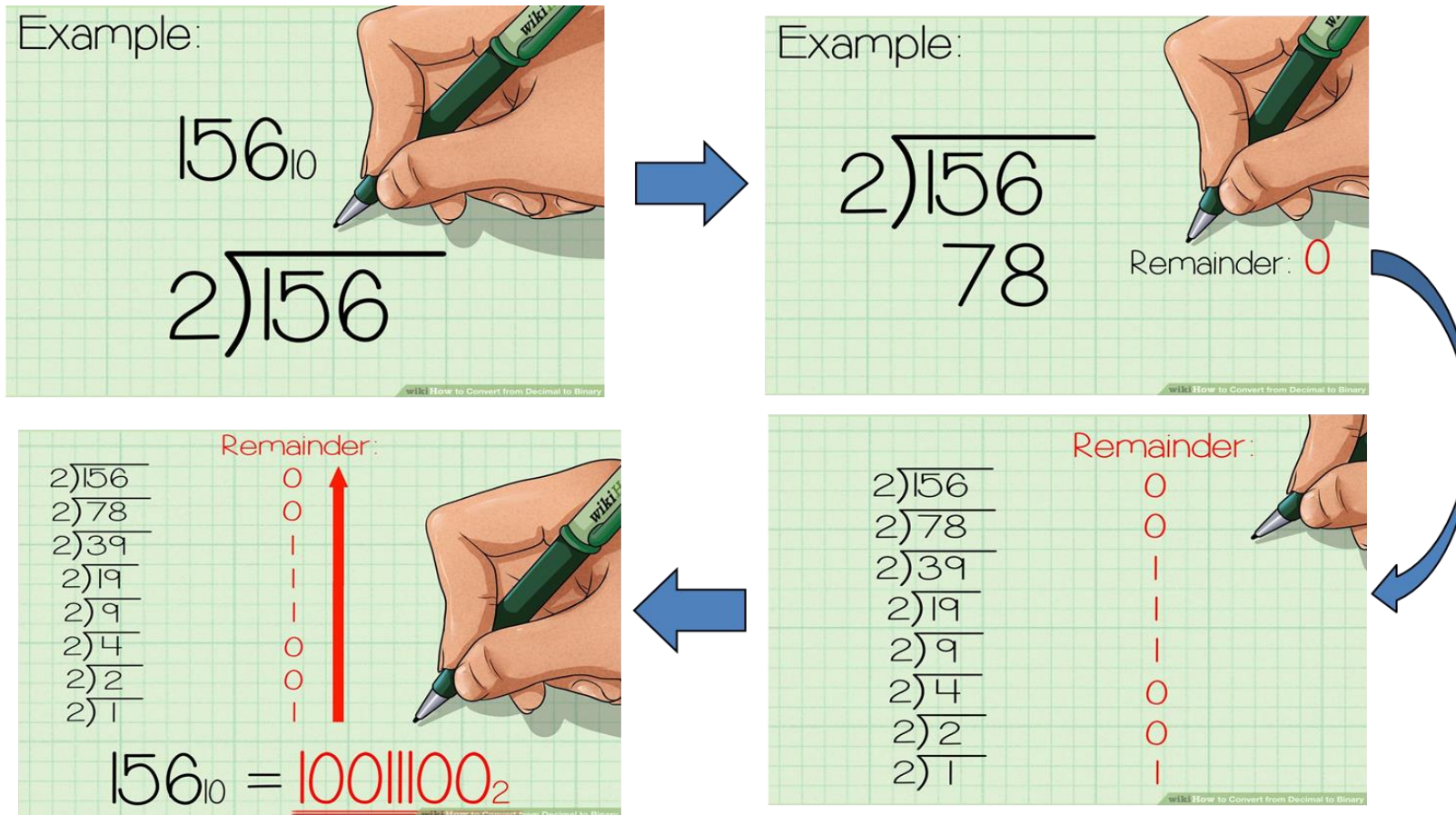
Answer

a)  $124_8 =$   
 $1 \times 8^2 + 2 \times 8^1 + 4 \times 8^0 = 64 + 16 + 4 = 84_{10}$

a)  $276_8 =$   
 $2 \times 8^2 + 7 \times 8^1 + 6 \times 8^0 = 128 + 56 + 6 = 190_{10}$

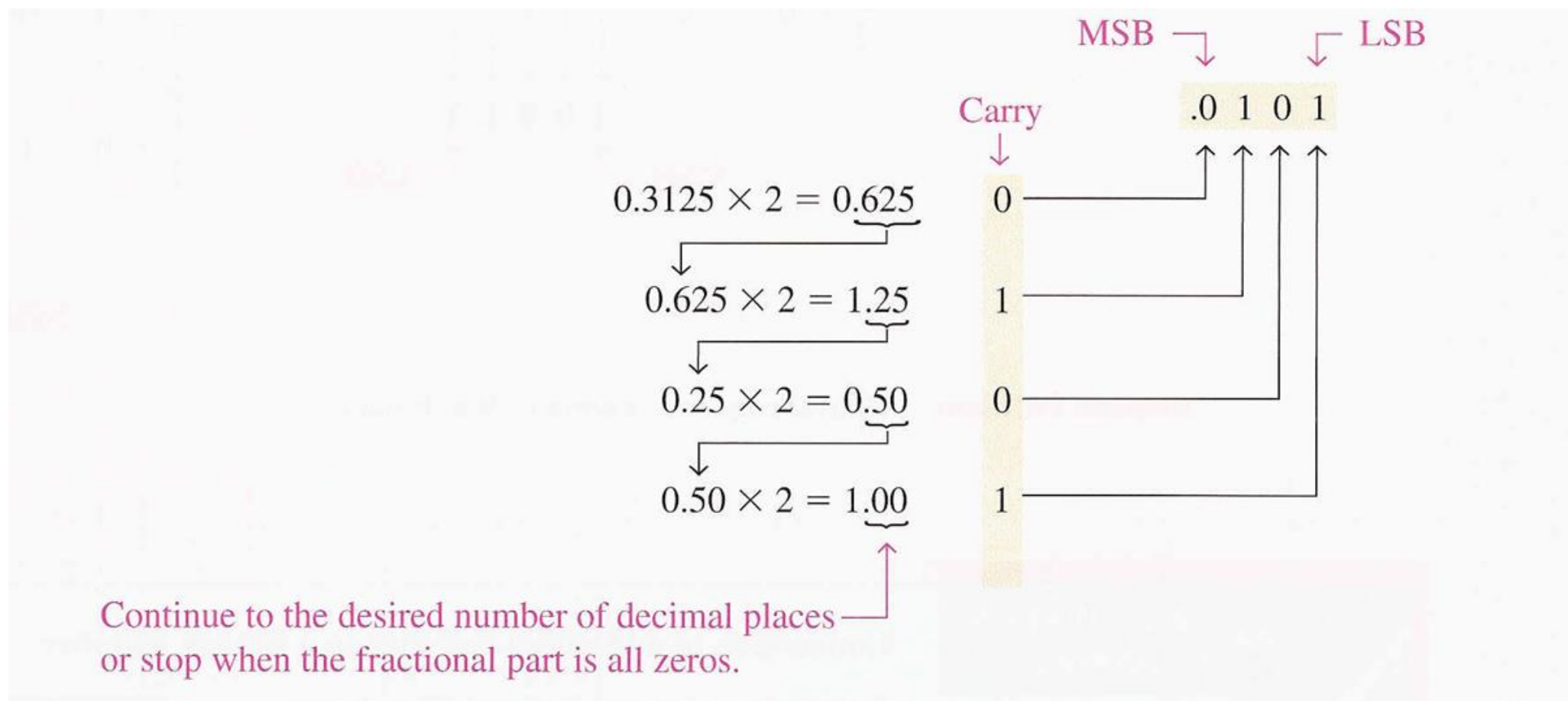
# Convert Decimal (Base 10) to Base N

Example 5: Convert the decimal number 156 to binary



# Convert Decimal (Base 10) to Base N

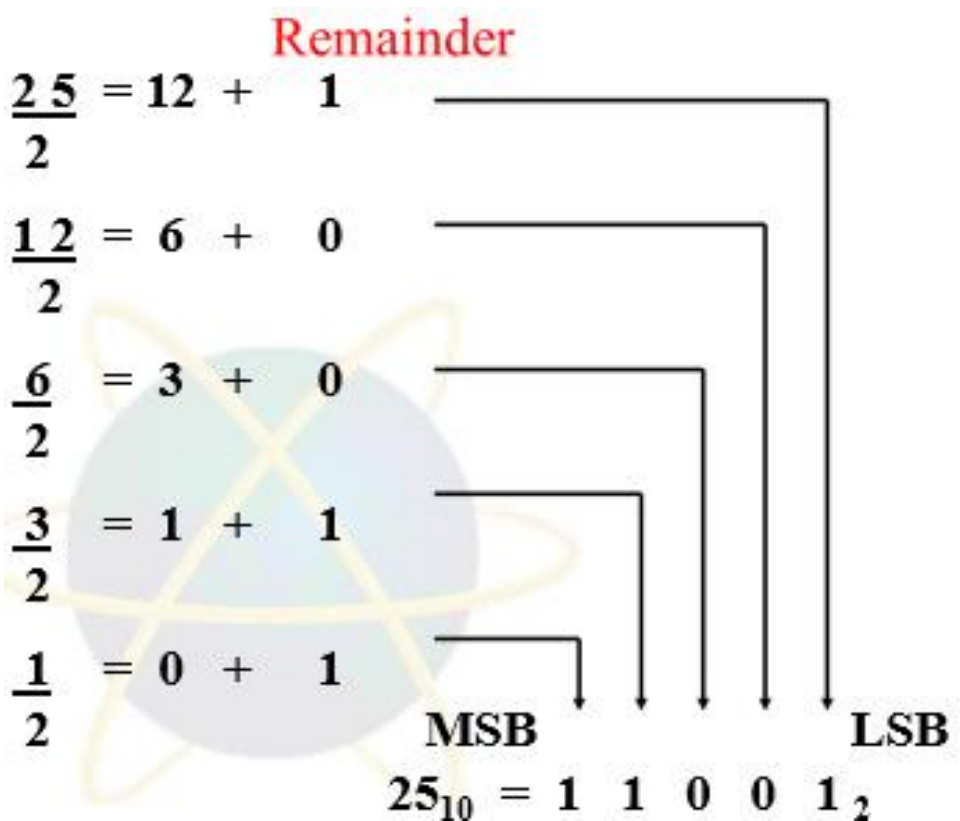
Example 6: Convert the decimal fractions 0.3125 to binary



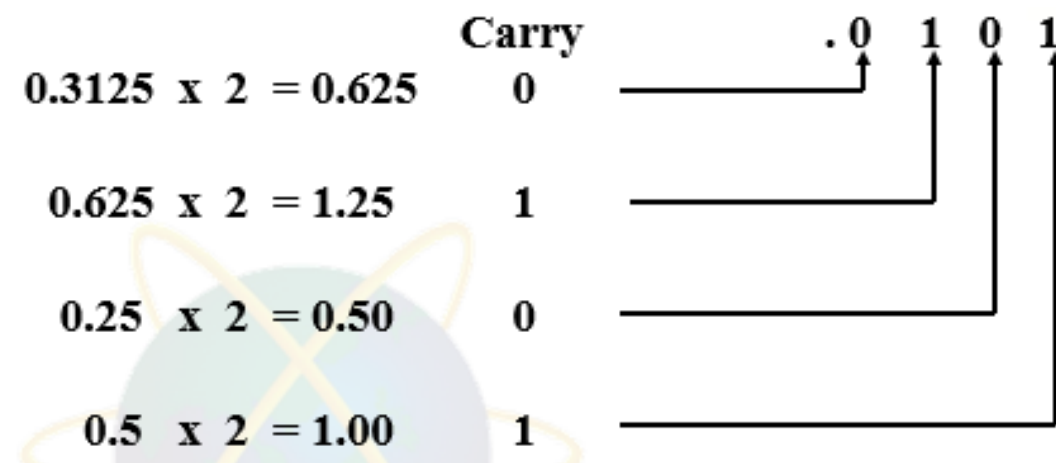
# Convert Decimal (Base 10) to Base N

Example 7: Convert the decimal fractions 25.3125 to binary

Step 1



Step 2



Answer:  $11001.0101_2$

# Convert Decimal (Base 10) to Base N

Example 8: Convert the following decimal to hexadecimal

- a)  $84_{10}$
- b)  $190_{10}$

Repeated division of a decimal by 16 will produce the equivalent hexadecimal number

Answer

a)

$$\begin{array}{r|l} 16 & 84 \\ 16 & 5 \longrightarrow 4 \uparrow \\ & 0 \longrightarrow 5 \end{array}$$

$54_{16}$

b)

$$\begin{array}{r|l} 16 & 190 \\ 16 & 11 \longrightarrow 14 = E \uparrow \\ & 0 \longrightarrow 11 = B \end{array}$$

$BE_{16}$

Example 9: Convert the following decimal to octal

- a)  $84_{10}$
- b)  $190_{10}$

Repeated division of a decimal by 8 will produce the equivalent octal number

Answer

a)

$$\begin{array}{r|l} 8 & 84 \\ 8 & 10 \longrightarrow 4 \uparrow \\ 8 & 1 \longrightarrow 2 \uparrow \\ & 0 \longrightarrow 1 \end{array}$$

$124_8$

b)

$$\begin{array}{r|l} 8 & 276 \\ 8 & 34 \longrightarrow 4 \uparrow \\ 8 & 4 \longrightarrow 2 \uparrow \\ & 0 \longrightarrow 4 \end{array}$$

$424_8$

# Convert Binary $\longleftrightarrow$ Hexadecimal

Example 10: Convert the following binary numbers to hex:

- a)  $1100101001010111_2$
- b)  $111111000101101001_2$

Convert, each 4-bit group to the equivalent hex, starting at the most right bit

Answer

(a)  $\underbrace{1100}_{C} \underbrace{1010}_{A} \underbrace{0101}_{5} \underbrace{0111}_{7} = CA57$

(b)  $\boxed{00}111111000101101001 = 3F169$

Two zeros have been added in part (b) to complete a 4-bit group at the left.

Example 11: Convert the following hex numbers to binary:

- a)  $10A4_{16}$
- b)  $23.A3_{16}$

Reverse the process. Convert each hex digit into 4 digit binary number

Answer

(a)  $10A4_{16}$

$1 \rightarrow 0001$   
 $0 \rightarrow 0000$   
 $A \rightarrow 1010$   
 $4 \rightarrow 0100$

$1001000010100100$

$23.A3_{16}$

$2 \rightarrow 0010$   
 $3 \rightarrow 0011$   
 $A \rightarrow 1010$   
 $3 \rightarrow 0011$

$0010\ 0011.1010\ 0011$

# Convert Binary $\longleftrightarrow$ Octal

Example 12: Convert the following binary numbers to octal:

- a)  $1100101001010110_2$
- b)  $1111.0001011011_2$

Convert, each 3-bit group to the equivalent hex, starting at the most right bit

Answer

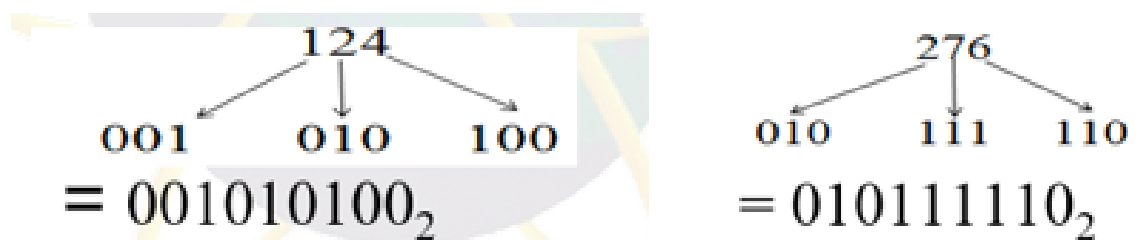
- a)  $\underbrace{1}_{1}\underbrace{100}_{4}\underbrace{101}_{5}\underbrace{001}_{1}\underbrace{010}_{2}\underbrace{110}_{6}$
- b)  $\underbrace{1}_{1}\underbrace{111}_{7}\underbrace{.000}_{0}\underbrace{101}_{5}\underbrace{101}_{5}\underbrace{100}_{4}$

Example 13: Convert the following octal numbers to binary:

- a)  $124_8$
- b)  $276_8$

Reverse the process. Convert each octal digit into 3 digit binary number

Answer



# Exercise: Convert the base number

|                                |                 |
|--------------------------------|-----------------|
| Convert binary to decimal      | $1110011_2$     |
|                                | $1110010.111_2$ |
| Convert decimal to binary      | $216_{10}$      |
|                                | $21.075_{10}$   |
| Convert hexadecimal to decimal | $10.01_{16}$    |
| Convert decimal to hexadecimal | $123.5_{10}$    |
| Convert octal to decimal       | $725_8$         |
| Convert decimal to octal       | $725.5_{10}$    |
| Convert Binary to hexadecimal  | $10000.1_2$     |
| Convert Hexadecimal to Binary  | $1F.C_{16}$     |
| Convert Binary to Octal        | $1001.1111_2$   |
| Convert Octal to Binary        | $47.3_8$        |

# Convert Base N to Base M

- Requires an intermediate conversion step
  - First convert base N to base 10
  - Then base 10 to base M.

Example 13: Convert  $18_9$  to  $?_{11}$

Answer:

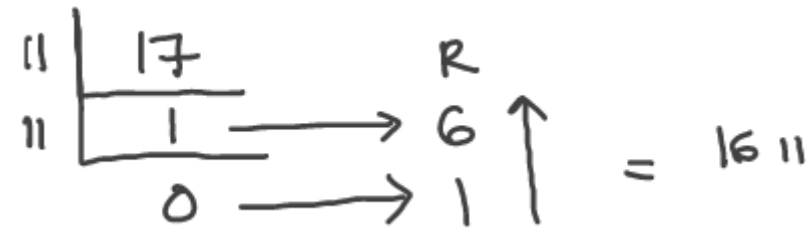
Step 1

Convert  $18_9$  to base 10:

$$\begin{aligned}
 18_9 &= (1 \times 9^1) + (8 \times 9^0) \\
 &= 9 + 8 \\
 &= 17_{10}
 \end{aligned}$$

Step 2

Convert from base 10 to base 11 using the divide-by-radix method:



Handwritten calculation showing the division of 17 by 11 to convert to base 11. The dividend 17 is written above a horizontal line, with 11 written to the left. The quotient 1 is written below the line, and the remainder 6 is written to the right. An arrow points from the 6 to the next step, which shows the remainder 6 divided by 11, resulting in a quotient of 0 and a remainder of 6. The final result is 16 in base 11, written as  $16_{11}$ .

Therefore  $18_9 = 16_{11}$

# Convert Base N to Base M

Example 14 : Convert  $AF_{16}$  to  $?_8$

Answer:

Step 1

Convert  $AF_{16}$  to base 10:

$$\begin{aligned} AF_{16} &= (10 \times 16^1) + (15 \times 16^0) \\ &= 160 + 15 \\ &= 175_{10} \end{aligned}$$

Step 2

Convert base 10 to base 8:

$$\begin{array}{r|l} 8 & 175 \\ \hline & 21 \\ 8 & 2 \\ \hline & 0 \end{array} \begin{array}{l} \longrightarrow R \\ \longrightarrow 7 \\ \longrightarrow 5 \\ \longrightarrow 2 \end{array} \uparrow = 257_8$$

Therefore  $AF_{16} = 257_8$

Example 14 : Convert  $AF_{16}$  to  $?_8$

Answer:

Step 1

Convert  $AF_{16}$  to base 2 (split into 4 bits binary):

$$AF_{16} = 10101111_2$$

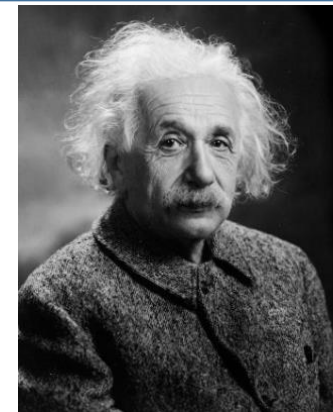
Step 2

Convert base 2 to base 8 (group into 3 bits binary):

$$\underbrace{010}_2 \underbrace{101}_5 \underbrace{111}_7_2$$

Therefore  $AF_{16} = 257_8$

Q & A  
Thank you



“Creativity is  
intelligence  
having fun”  
– Albert Einstein